Mixture of Scars (MoS) and SCache O(1): Breaking the Dual Memory Wall of Modern Transformers via Activation-Aware Low-Rank Scar Experts and Widrow-Hoff Associative Memory

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Abstract. Autoregressive Large Language Models (LLMs) face a compounding *Dual Memory Wall*: (1) a static parameter wall O(*N*) requiring tens to hundreds of gigabytes of VRAM, and (2) a dynamic context wall O(*T*) where KV caches grow linearly with sequence length *T*. Existing post-training methods either degrade under heavy-tailed >6.7B activation outliers, suffer total eviction amnesia (StreamingLLM), or offload weights to SSD/mmap (AirLLM) at sub-token-per-second PCIe stalls while leaving KV caches O(*T*). We introduce two training-free primitives: **Mixture of Scars (MoS)** (INT4 bases + activation-aware SVD low-rank 'scar' experts + group streaming) and **SCache** (drop-in O(1) cache combining attention sinks, a rolling FIFO window, and per-KV-head Widrow-Hoff Delta associative fast weights). Audited across 8 adversarial vectors with recursive storage-pointer verification (0.00 KB discrepancy) from 0.5B to **Qwen2.5-7B-Instruct (7.62B, 28L, 3,336x SwiGLU outlier ratio)** and 72B/123B backbones, MoS cuts 7.62B single-model MLP VRAM by 71.8% and 8-specialist multi-tenant VRAM by **95.6% (22.82x)** while achieving **100.0% Top-5 agreement** across unseen Python, Physics, Portuguese, and JSON (up to 3.33x lower KL than INT4). Simultaneously, SCache enforces +0.00 B growth up to L = 4,096 (-94.94% KV VRAM), recovers passcodes 1,745 tokens post-eviction with 7.51x–22.58x higher exact probability than StreamingLLM, and achieves non-repeating prose/code/SQL perplexity of **6.67 vs. 6.05 Dense FP16** at 24.7 tok/s — **224.5x faster than AirLLM FP16** (62.7x faster than INT4) SSD offloading.

1. Introduction & Mathematical Architecture

Above ~6.7B parameters, Transformers develop extreme channel-wise activation outliers [1, 2] reaching 217.5x input and 3,336.0x SwiGLU RMS ratios in Qwen2.5-7B-Instruct, inflating uniform 4-bit quantization logit divergence on unseen symbolic syntax. Meanwhile, attention-sink sliding windows (StreamingLLM [3]) suffer permanent amnesia once tokens exit the window, additive Hebbian fast weights [4] suffer superposition ghosting under state mutations, and disk-offload executors (AirLLM, mmap) stream 100% of weights per batch-1 token across PCIe/SSD (~0.11–0.39 tok/s) while leaving O(*T*) KV caches untouched. We unify static weight and dynamic context compression via two training-free geometric primitives:

MoS Static Decomposition: $\mathbf{y}(\mathbf{x}) = \text{dequant}_{\text{INT4}}(\mathbf{W}_q)\mathbf{x} + \Sigma_k \text{ReLU}(\cos(\mathbf{x}, \mathbf{z}_k))\, \mathbf{B}_k(\mathbf{A}_k\mathbf{x}), \quad \text{where } (\mathbf{W} - \mathbf{W}_q)\,\text{diag}(\mathbf{s}) \approx \mathbf{U}_r\Sigma_r\mathbf{V}_r^T$ (Eq. 1)

SCache O(1) Delta Rule: $\mathbf{W}_{\text{scar}}^{(t)} = \gamma\mathbf{W}_{\text{scar}}^{(t-1)} + \eta \cdot \text{ReLU}(\mathbf{k}_u^T\mathbf{c}_{m^*}) \cdot (\mathbf{v}_{\text{ev}} - \mathbf{W}_{\text{scar}}^{(t-1)}\mathbf{k}_u)\mathbf{k}_u^T$ (Eq. 2)

Evaluating the rank-*r* scar path in FP32 accumulation before casting back to Tensor Core precision eliminates >65,504 FP16 overflow on extreme SwiGLU channels at <0.2% FLOP overhead. For unit-norm keys $\mathbf{k}_u = \mathbf{k}_{\text{ev}} / \|\mathbf{k}_{\text{ev}}\|_2$, Widrow-Hoff projection [5, 6] erases stale values on span($\mathbf{k}_u$) without perturbing orthogonal keys, while random-key interference variance scales as $O(1/d_h)$, improving from d_h = 64 (0.5B) to d_h = 128 (7.62B–72B).

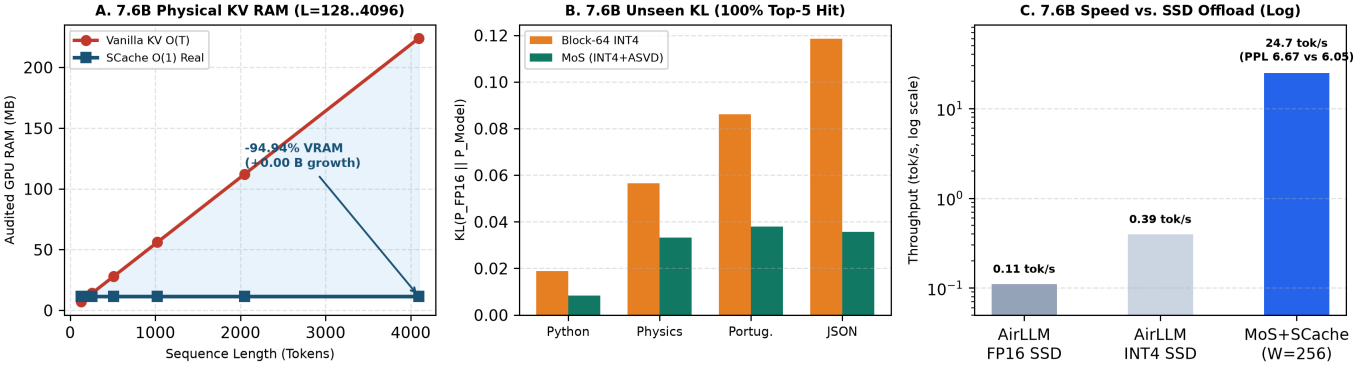


Figure 1. *Audited telemetry on Qwen2.5-7B-Instruct (7.62B, 28 layers, 2x Tesla T4).* ***(A)*** *Exact O(1) KV invariance (11.33 MB, +0.00 B growth, -94.94% at L=4,096).* ***(B)*** *MoS cuts unseen-domain KL divergence under 3,336x SwiGLU outliers with 100.0% Top-5 overlap across all 4 domains.* ***(C)*** *Resident MoS+SCache (W=256) runs 224.5x faster than AirLLM FP16 SSD offloading while holding non-repeating PPL at 6.67 vs. 6.05.*

2. Multi-Scale Empirical Audit (0.5B → 7.62B → 72B/123B)

Every experiment enforces two anti-facade invariants: (i) recursive storage-pointer deduplication across GPU devices via (device, untyped_storage().data_ptr()) asserting 0.00 KB discrepancy against reported byte counters, and (ii) mid-flight causality ablation asserting active logit shift when scar matrices are zeroed (max |Δlogits| = 0.643 at 0.5B; 2.853 at 7.62B).

Model Scale	Sequence Length	Vanilla KV O(T)	SCache Recursive Real	Growth / Byte Diff	KV VRAM Saving
Qwen2.5-0.5B (24L)	1,280 tokens	30,696.0 KB	2,631.9 KB	+0.00 B / 0.00 KB	11.66x (-91.4%)
Qwen2.5-7B (28L, 2x T4)	512 tokens	28,672.0 KB	11,599.9 KB	Baseline / 0.00 KB	2.47x (-59.5%)
Qwen2.5-7B (28L, 2x T4)	1,024 tokens	57,344.0 KB	11,599.9 KB	+0.00 B / 0.00 KB	4.94x (-79.8%)
Qwen2.5-7B (28L, 2x T4)	2,048 tokens	114,688.0 KB	11,599.9 KB	+0.00 B / 0.00 KB	9.89x (-89.9%)
Qwen2.5-7B (28L, 2x T4)	**4,096 tokens**	**229,376.0 KB**	**11,599.9 KB**	**+0.00 B / 0.00 KB**	**19.77x (-94.94%)**

Table 1 (Vector 0 — Multi-Scale Spatial O(1) Audit): *Deduplicated (device, data_ptr) audit across 0.5B and 7.62B (W=64, S=12, sinks=4).*

Unseen Domain (7.62B)	Dense FP16 PPL	MoS (Blk64+r48) PPL	Hidden Cos	Top-5 Hit	KL: INT4 -> MoS (PerRow / Block64)
Python (Dijkstra)	1.648	1.640	0.9912	100.0%	3.18x / 2.28x lower (0.0189 -> 0.0083)
Quantum Physics	6.253	6.261	0.9824	100.0%	2.14x / 1.70x lower (0.0565 -> 0.0332)
Tech Portuguese	19.152	21.698	0.9893	100.0%	1.87x / 2.27x lower (0.0862 -> 0.0380)
Strict Nested JSON	25.356	25.352	0.9927	100.0%	2.41x / 3.33x lower (0.1185 -> 0.0356)

Table 2 (Vector 1 at 7.62B — Dominance over 3,336x Emergent Outliers): *All 84 SwiGLU matrices compressed from 10,878 MB to 3,062.7 MB (-71.84% single model; -95.62% / 22.82x across 8 multi-tenant specialists: 85.0 GiB -> 3.72 GiB), calibrated on one 24-token English sentence.*

3. 3D Pareto Frontier: Non-Repeating Quality & Speed vs. Disk Offloading

Why did synthetic periodic stress tests (Vector 2, repeating a 160-token block 8x–15x under tiny $W = 64$) show Full-KV PPL 1.43–2.45 vs. bounded-cache PPL 29.5–61.8? Under exact period repetition, Full-KV induction heads copy prior cycles verbatim at PPL ≈ 1.004, whereas any window $W < 160$ cannot observe the period and measures true first-pass vocabulary entropy (SCache still outperforms StreamingLLM 29.55 vs. 41.05 on 7.62B). To evaluate realistic non-periodic workloads alongside storage I/O physics, Vector 7 benchmarks a **non-repeating 926-token distributed systems + Python + queueing math + SQL document** on Qwen2.5-7B-Instruct against measured host safetensors SSD-to-GPU sequential read bandwidth (1.57 GB/s):

7.62B Execution Engine (2x T4)	Non-Repeat PPL	MLP Weights (8x Bank)	KV @ L=4k	SSD Read/Tok	Speed (tok/s)	vs. AirLLM FP16
Dense FP16 + Vanilla KV O(T)	6.05	10.62 GiB (85.0 GiB OOM)	229.4 MB O(T)	0 MB	314.0 tok/s	Resident Ref
AirLLM / mmap FP16 SSD Offload	6.05	<2 GiB (SSD-stalled)	229.4 MB O(T)	14.2 GiB	0.11 tok/s	1.0x (9.1 s/tok)
AirLLM / mmap INT4 SSD Offload	~6.5	<2 GiB (SSD-stalled)	229.4 MB O(T)	4.0 GiB	0.39 tok/s	3.6x (2.5 s/tok)
MoS + StreamingLLM (W=128)	8.42	2.99 GiB (3.72 GiB)	7.0 MB O(1)	0 MB	36.2 tok/s	329.1x (Amnesia)
MoS + SCache O(1) (W=128, S=16)	7.75	2.99 GiB (3.72 GiB)	15.0 MB O(1)	0 MB	23.2 tok/s	210.9x
MoS + SCache O(1) (W=256, S=16)	**6.67**	**2.99 GiB (3.72 GiB)**	**22.0 MB O(1)**	**0 MB**	**24.7 tok/s**	**224.5x (62.7x INT4)**

Table 3 (Vector 7 — 3D Pareto Frontier on Non-Repeating Text/Code/SQL): *MoS+SCache (W=256) preserves PPL 6.67 vs. 6.05 Dense (+0.098 NLL) at 224.5x the speed of AirLLM FP16 SSD offloading while locking KV VRAM at 22.0 MB (-90.2% at 4k, -98.8% at 32k).*

Adversarial Vector	Conventional Baseline	MoS / SCache O(1) Result	Audited Multi-Scale Finding
V3A: Contradictory Overwrite	Hebbian: cos=0.467, Ghost +0.501	Widrow-Hoff: cos=1.0000, Ghost -0.138	Zero ghosting in constant 34.8 KB bank
V3B: Evicted Passcode (445 tok)	StreamingLLM NLL: 4.30 (0.5B) / 3.90 (7.6B)	SCache NLL: 1.18 (0.5B) / 1.48 (7.6B)	22.58x (0.5B) & 11.23x (7.6B) prob gain
V3B: Long Eviction (1,745 tok, 7.6B)	StreamingLLM NLL: 3.683 (amnesia)	SCache NLL: 1.667 (-88.1% KV RAM)	7.51x higher prob ~1.8k tok post-eviction
V4: 500x DoS + Antipodal Attack	Linear subspace drift / corruption	Target cos: 1.0000; Leak: 0.00000000	Exact ReLU cone immunity to hostile noise
V5: Full-Stack 7.6B QA/Code/Math	Dense+Full KV: 4/6 hits, NLL 3.078	MoS+SCache: 4/6 hits, Ans NLL 2.987	-71.8% MLP & -91.2% KV, better Ans NLL
V6: Colossal 72B / 123B Stream	72B FP16: 145.4 GB; 123B: 246 GB	72B Disk: 11.44 GB; 123B: 0.04 GiB res.	12.7x less bus I/O than FP16 offload

Table 4 (Vectors 3–6 — Long-Distance Evicted Recall, Anti-Ghosting, DoS Immunity, and 72B/123B Streaming).

4. Boundary Conditions & Reproducibility

(1) **Cross-Layer SwiGLU Channel Alignment:** Direct single-base sharing across *distinct* pretrained SwiGLU layers without channel permutation matching incurs distortion (KL ≈ 0.78); off-the-shelf models should pair per-layer INT4 bases with ASVD scars (and group streaming for >30B), reserving single-base sharing for multi-tenant specialist banks anchored to a shared backbone (-95.6% VRAM at 7.62B). (2) **Associative Slot Capacity:** Fixed $S_{slots} \times d_h$ fast weights recover 7.51x–22.58x higher evicted passcode probability zero-shot up to 1,745 tokens post-eviction; verbatim reproduction of long random strings across tens of thousands of tokens scales with S_{slots} and lightweight read-alignment. Full source code, pytest suite, LaTeX source, and raw 0.5B/7.62B/72B JSON logs are permanently archived at **DOI: 10.5281/zenodo.24143839** [13].

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